Exploring Image-to-Image Transformations: A Comparative Analysis

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***Abstract*—Image-to-image translation is an important task in computer vision, enabling applications like style transfer and object manipulation. CycleGAN, with its cycle consistency loss, made it possible to train models on unpaired datasets, and it has been a foundational approach despite newer models like StarGAN and StyleGAN offering improvements in scalability and realism.**

In this study, we explore ways to improve CycleGAN for unpaired gender transformation tasks. We added self-attention layers, spatial attention mechanisms, and U-Net modifications to enhance the quality of the generated images and maintain semantic consistency. Using a preprocessed UTKFace dataset, we evaluated these changes both quantitatively and qualitatively. Notably, the spatial attention mechanism performed better than self-attention and U-Net, achieving a lower FID score of 14.76 compared to 20.19 for Female $\rightarrow$ Male transformations.

Interestingly, transformations from Female to Male consistently performed better than Male to Female across all models, possibly because of the greater challenges in capturing subtle feminine traits. StarGANv2 demonstrated stronger overall performance, particularly due to its superior style control and its ability to achieve these results while training for only 1.875% of the images required by the other models. However, this efficiency came at the cost of longer training durations due to StarGAN v2's architectural complexity. In contrast, our simplified modifications to CycleGAN offer a promising alternative, achieving competitive results with fewer parameters and significantly reduced runtime. This work underscores the adaptability of foundational models like CycleGAN, demonstrating how targeted architectural changes can address modern challenges in image-to-image translation.

I. Introduction

Image-to-image translation is a pivotal task in computer vision, enabling various transformations across domains, such as style transfer, object manipulation, and domain adaptation. The advancement in this area is largely due to generative adversarial networks (GANs). With their ability to learn complex mappings between two image domains, GANs provide frameworks capable of generating high-quality outputs with minimal data requirements. CycleGAN in particular emerged as a foundational model. Through the introduction of cycle consistency loss, CycleGAN enabled unpaired image-to-image translation. This marked a significant step forward, as it allowed models to operate on datasets lacking explicit correspondences between source and target domains.

Following CycleGAN, subsequent developments in GAN architecture such as StarGAN and StyleGAN introduced innovations that expanded the scope and quality of GAN-based methods. StarGAN enabled multi-domain transformations within a single framework [1], while StyleGAN focused on producing photorealistic outputs with greater control over image synthesis [2]. Despite these advancements, CycleGAN remains influential due to its simplicity and effectiveness, making it an ideal candidate for studying how architectural modifications can improve unpaired image translation.

In this study, we propose modifications to CycleGAN's architecture, including the integration of self-attention layers, to evaluate their impact on translation quality. We selected CycleGAN for this exploration due to its historical significance and adaptability. Our goal is to assess whether these changes can bridge the gap between CycleGAN and more recent state-of-the-art models. Specifically, we applied these modifications to the task of gender transformation, a domain well-suited to CycleGAN's strengths given the lack of paired datasets. This domain is meaningful in unpaired image-to-image translation because it involves altering key facial features, like the jawline, makeup, or smoothness of the face, while still preserving the person's unique identity. Achieving a balance between subtle changes and maintaining identity makes this domain an excellent way to test and refine model improvements. By formulating hypotheses on how architectural changes affect outcomes and rigorously evaluating the results, we aim to contribute insights into the evolution of neural network design for unpaired image translation.

II. Background

A. CycleGAN

CycleGAN [3] was a landmark in the field of unpaired image-to-image translation. The model's central innovation was the use of cycle consistency loss, which enforces that an image translated from one domain to another and back retains its original characteristics. This approach obviated the need for paired datasets, making it applicable to a wide range of real-world tasks where paired examples are unavailable.

CycleGAN employs two generator-discriminator pairs: one for mapping images from domain X to domain Y and another for the reverse direction. The cycle consistency loss, combined with adversarial loss, encourages the generators to produce realistic outputs while preserving semantic consistency. Despite its effectiveness, CycleGAN has limitations, including difficulty in capturing fine-grained details and a lack of scalability to multi-domain tasks, motivating subsequent research into architectural enhancements.

B. StarGAN and StarGANv2

StarGAN [1] addressed the scalability issue by introducing a single model capable of performing multi-domain image translation. Using domain-specific attributes as inputs, StarGAN learned mappings across multiple domains simultaneously, significantly reducing the computational cost of training separate models for each domain. Its successor, StarGANv2 [4], extended this approach by introducing style vectors sampled from learned distributions. This addition enhanced the diversity and realism of generated images, making StarGANv2 one of the most versatile frameworks for multi-domain image translation.

C. StyleGAN and StyleGANv2

StyleGAN [2] shifted the focus from image translation to high-quality image synthesis. By decoupling the latent space from the image synthesis process, StyleGAN introduced fine-grained control over image attributes, achieving unprecedented levels of photorealism. StyleGANv2 [5] refined this approach, addressing issues such as training instability and visual artifacts, and became a benchmark for generative modeling. While primarily designed for image synthesis, StyleGAN's innovations have inspired enhancements in image translation models, particularly in controlling the diversity and quality of outputs.

D. Other Models

Several other models have emerged to address specific challenges in image translation. CUT (Contrastive Unpaired Translation) [6] leverages contrastive learning to improve feature representation and translation quality, demonstrating strong performance in preserving content structure. FUNIT (Few-Shot Unsupervised Image Translation) [7] tackles the challenge of domain adaptation with limited examples, achieving impressive results in few-shot learning scenarios. Additionally, there are other models [8], [9] that have contributed to the field by focusing on high-resolution outputs and multimodal diversity, respectively.

These advancements highlight the dynamic evolution of image-to-image translation techniques. By revisiting CycleGAN and introducing targeted modifications, our study seeks to explore whether its foundational architecture can be adapted to remain competitive with more recent approaches. The focus on gender transformation as a use case allows us to assess the practical implications of these modifications in a domain that epitomizes the challenges and opportunities of unpaired image translation.

III. Models and Ablation Study

In this section, we describe the base CycleGAN model and detail the architectural modifications implemented to enhance its performance. These modifications are motivated by the need to improve image quality and address specific limitations of the original CycleGAN model in the context of unpaired image-to-image translation tasks, specifically in the context of gender transformation.

A. Preprocessing Pipeline

Our preprocssing pipeline consisted of four critical steps to prepare the dataset and ensure high-quality inputs for the models.

1) Dataset and Domain Separation: The UTKFace [10] dataset, which consists of face images with annotations of age, gender, and ethnicity in file names. Metadata parsing was automated to separate the dataset into two primary domains: ”M” for males and ”F” for females. This process facilitated clear domain definition for training and evaluation.

2) Face Cropping: To eliminate extraneous background information and emphasize domain-relevant features, we employed the Multi-Task Cascaded Convolutional Neural Network (MTCNN) [11] for face detection and cropping. The MTCNN model identified facial bounding boxes, which were used to crop the images around facial regions. The face detection process leveraged GPU acceleration, ensuring faster processing times.

3) Quality Assurance and Dataset Refinement: Manual inspection of the cropped dataset revealed several issues, including low resolution, indistinct facial features, and color artifacts, all of which could adversely affect model training. To address these issues, we implemented a rigorous human review process to ensure dataset quality. Each domain (”M” for males and ”F” for females) was independently inspected by separate researchers. To validate the reliability of this process, the researchers cross-reviewed 10% of the dataset from the opposite domain. This inter-coder reliability assessment achieved a Cohen's Kappa score of 1, indicating perfect agreement between the annotators.

In addition to quality control, we ensured demographic balance within the dataset by stratifying samples across age, ethnicity, and gender ratios. Elderly populations were excluded to mitigate confounding factors unrelated to the primary focus of gender transformation. These efforts resulted in a high-quality, diverse, and well-balanced dataset optimized for model training and evaluation.

Female Domain

The original dataset contained 12,091 samples. Following filtration, 7,940 samples were retained, with the remaining data removed due to:

- Items out of the age range: 3,487 samples
- Flawed data (e.g., low resolution or artifacts): 624 samples
- Male samples incorrectly labeled as female: 40 samples

Male Domain

The original dataset contained 12,717 samples. After filtering, 7,624 samples were retained. Removed data included:

- Items out of the age range: 3,693 samples
- Flawed data (e.g., low resolution or artifacts): 1,278 samples
- Female samples incorrectly labeled as male: 122 samples

4) Image Resizing and Dataset Splitting: All images were resized to 256×256 pixels for uniformity. The dataset was then split into training and validation subsets using an 80/20 split ratio. Random shuffling within each domain ensured well-balanced and unbiased partitions, maintaining robust data distribution for training and evaluation.

B. CycleGAN Base

The baseline for our experiments is the CycleGAN model introduced by Zhu et al. [3]. CycleGAN performs unpaired image-to-image translation by leveraging two primary losses: the adversarial loss, which ensures that the generated images are indistinguishable from real ones, and the cycle consistency loss, which enforces that translating an image to the target domain and back to the original domain preserves the image's content. The generator architecture in CycleGAN uses ResNet blocks, while the discriminator employs PatchGANs.

For our baseline model, we retained the original architecture. The baseline model serves as a reference point for evaluating the impact of our architectural modifications.

C. CycleGAN Enhanced

To address the limitations of the base CycleGAN model, we introduced three major architectural enhancements: self-attention, spatial attention, and U-Net modifications to the generator. We train all models for 100 epochs with a learning rate of 0.001 and apply the same linear decaying strategy over the next 100 epochs on an NVIDIA A100 GPU in the Google Colaboratory setting.

1) CycleGAN w/ Self-attention Mechanism:

- **Motivation:** The self-attention mechanism allows the model to capture long-range dependencies within an image by enabling each pixel to attend to every other pixel. This is particularly advantageous for gender transformation tasks, where changes in one region (e.g., facial structure) must align harmoniously with distant regions (e.g., hairstyle or jawline). Such global consistency is crucial for producing realistic and coherent results.
- **Implementation:** The self-attention mechanism was incorporated into the generator architecture by adding a self-attention layer to key residual blocks in the ResNet backbone. This layer computes attention maps that guide the generator in focusing on spatial relationships between different regions of the image. Training takes about 10 hours on an NVIDIA A100 GPU.
- **Expected Benefits:** By explicitly modeling spatial dependencies, the self-attention mechanism improves the semantic coherence of transformations across regions. For instance, when softening facial features for Male $\rightarrow$ Female transformations, the model ensures that modifications to the eyes, nose, and lips align with broader structural adjustments, such as the jawline and hairstyle. This leads to sharper and more realistic outputs while mitigating artifacts.

2) CycleGAN w/ Spatial-attention Mechanism:

- **Motivation:** Spatial attention focuses on the spatial regions of an image that are most important for a transformation. Instead of considering relationships between all features, as self-attention does, spatial attention prioritizes regions in the height and width dimensions, emphasizing the locations the model should pay attention to.
- **Implementation:** The implementation is the same as for self-attention, just replacing the self-attention layer with a spatial one. Training takes about 10 hours on an NVIDIA A100 GPU.
- **Expected Benefits:** Since gender translation often involves transforming specific features (e.g. eyes, lips, jawline) without drastically altering the rest of the image, spatial attention can help focus on these key areas. As a result, spatial attention will likely better handle subtle details like facial contours or makeup effects, which are critical for perceptual quality in our gender translation task. Spatial attention is also computationally cheaper than self-attention since it doesn't calculate pairwise relationships across all spatial locations.

3) CycleGAN w/ different architecture changes to U-Net:

- **Motivation:** The original CycleGAN architecture employs fixed hyperparameters that may not optimally balance model capacity and computational efficiency. In the context of gender transformation, the network must capture both fine facial details and broad structural changes, suggesting that a careful rebalancing of network capacity parameters could improve performance without increasing computational overhead.
- **Implementation:** Our implementation focuses on strategic modifications to two key capacity parameters in the generator architecture. We reduced the number of generator filters (ngf) from 64 to 48 while maintaining discriminator filters (ndf) at 64, and increased the number of residual blocks from 9 to 12. This approach maintains the fundamental CycleGAN structure while optimizing its capacity distribution for gender transformation tasks. By reducing generator base filters to 48, we decrease the parameter count in feature extraction layers, while the increased number of ResNet blocks provides deeper feature processing to compensate. The discriminator's capacity remains unchanged at 64 filters to ensure robust adversarial feedback. The model was trained for 100 epochs on an NVIDIA A100 GPU.
- **Expected Benefits:** These architectural adjustments results in a less overall parameter count by 26.2% (11.4M vs 8.4M) and improved distribution for the specific task of gender transformation. The deeper network structure, en-

abled by additional residual blocks, enhances the model's ability to learn complex gender-specific features, while the reduced filter count improves memory efficiency. The maintained discriminator capacity ensures stable training through strong adversarial signals. This optimization particularly targets the challenge of gender transformation, where both fine detail preservation and efficient feature processing are crucial for convincing results.

D. StarGANv2

The baseline for this segment of our experiments is the StarGAN v2 model, utilized in its original form as introduced by Choi et al. [4]. StarGAN v2 is designed for multi-domain image-to-image translation, leveraging style-based generators and adaptive instance normalization (AdaIN) to achieve high-quality transformations between multiple domains.

StarGAN v2 includes a **style encoder** that extracts style information from reference images, helping the generator create outputs that match the look of the reference. This feature allows for precise control over the appearance of the generated images, making it useful for detailed style adjustments. The model can also produce different styles within the same domain, making it more flexible and adaptable.

For our experiments, we worked with two domains: male and female. StarGAN v2's ability to leverage reference images allowed the model to not only transform images between domains but also incorporate stylistic features from the referenced individual. While this capability added an additional layer of flexibility—allowing for variations such as changes in hair texture or makeup—it was not the primary focus of our study. Instead, we utilized StarGAN v2 as a control to evaluate the ideal performance of a proven method for domain transformations.

We used StarGAN v2 without any architectural modifications, training the model for 15,000 iterations with a batch size of 4 images per iteration on an NVIDIA A100 GPU. By employing the model in its default configuration, we ensured a robust baseline for comparing its domain transformation capabilities with other models and methods explored in this study. However, after accounting for batch size and dataset size differences, this model saw 1.875% of the images that the other models saw for training, highlighting the relative efficiency of StarGAN v2 in achieving comparable results with significantly fewer training iterations. Nevertheless, the training duration was comparable to that of the full dataset processed by the spatial CycleGAN, due to the higher architectural complexity of StarGAN v2.

IV. Evaluation

A. Quantitative

The key observations from the quantitative results are shown in Figure 1:

1) Base CycleGAN: The baseline CycleGAN model yielded acceptable results for unpaired gender transformation tasks but revealed several limitations. These results serve as a benchmark for evaluating the impact of the architectural modifications introduced in the enhanced versions of the model.

- **Cycle Consistency Error:** M $\rightarrow$ F and F $\rightarrow$ M exhibit nearly identical, low cycle consistency errors (0.0751 and 0.0752 respectively).
- **Identity Preservation Error:** The identity preservation errors are slightly lower for the Female $\rightarrow$ Male transformation (0.0669) compared to Male $\rightarrow$ Female (0.0702). This suggests that the model retains facial identity better when performing Female $\rightarrow$ Male transformations.
- **Visual Realism (FID Scores):** The FID scores indicate an equal level of transformation quality for both directions. The Female $\rightarrow$ Male transformation FID score (16.44) is only slightly lower than the Male $\rightarrow$ Female (16.87) FID score.

Overall Performance: Although the base CycleGAN successfully performed unpaired gender transformation, its outputs often appeared less natural and consistent compared to those from enhanced models.

2) CycleGAN w/ Self-attention Mechanism: The key observations from the quantitative results are as follows:

- **Cycle Consistency Error:** Both Male $\rightarrow$ Female and Female $\rightarrow$ Male transformations exhibit low cycle consistency errors for the self attention mechanism (0.0735 and 0.0742, respectively). This demonstrates the model's ability to reconstruct the original image accurately when transforming back from the target domain.
- **Identity Preservation Error:** The identity preservation errors are slightly lower for the Female $\rightarrow$ Male transformation (0.0756) compared to Male $\rightarrow$ Female (0.0763). This indicates that the model still retains facial identity better when performing Female $\rightarrow$ Male transformations.
- **Fréchet Inception Distance (FID):** The FID scores highlight a clear difference in transformation quality. The Female $\rightarrow$ Male transformation achieves a significantly lower FID score (20.19) compared to Male $\rightarrow$ Female (27.32). This indicates that Female $\rightarrow$ Male transformations produce more visually realistic and domain-consistent images.

3) CycleGAN w/ Spatial-attention Mechanism: The key observations from the quantitative results are as follows:

- **Cycle Consistency Error:** Both Male $\rightarrow$ Female and Female $\rightarrow$ Male transformations exhibit low cycle consistency errors for the spatial attention mechanism (0.0681 and 0.0648, respectively). This demonstrates the model's ability to reconstruct the original image accurately when transforming back from the target domain.
- **Identity Preservation Error:** In contrast to the self-attention extension, the identity preservation errors are slightly higher for the Female $\rightarrow$ Male transformation (0.0629) compared to Male $\rightarrow$ Female (0.0616). This suggests that the model retains facial identity better when performing Male $\rightarrow$ Female transformations.
- **Fréchet Inception Distance (FID):** The FID scores show a difference in transformation quality in the same

Metric	Base	w/ Self-Attention	w/ Spatial Attention	w/ U-Net Modifications
Cycle Consistency Error (M → F)	0.0751	0.0735	0.0681	0.0678
Cycle Consistency Error (F → M)	0.0752	0.0742	0.0648	0.0646
Identity Preservation Error (M → F)	0.0702	0.0763	0.0616	0.0649
Identity Preservation Error (F → M)	0.0669	0.0756	0.0629	0.0661
FID Score (M → F)	16.87	27.32	16.48	21.73
FID Score (F → M)	16.44	20.19	14.76	15.13

Fig. 1: Evaluation Metrics for Gender Transformation using Different Models

direction as the self-attention extension. The Male → Female transformation achieves a higher FID score (16.48) compared to Female → Male (14.76). This indicates that Female → Male transformations produce more visually realistic and domain-consistent images.

4) CycleGAN w/ U-Net Modification: The key observations from the quantitative results are as follows:

- **Cycle Consistency Error:** Both Male → Female and Female → Male transformations exhibit lower cycle consistency errors than the spatial attention mechanism (0.0678 and 0.0646, respectively).
- **Identity Preservation Error:** Both Male → Female and Female → Male transformations exhibit higher cycle consistency errors than the spatial attention mechanism (0.0649 and 0.0661, respectively).
- **Fréchet Inception Distance (FID):** The Female → Male transformation achieves a significantly lower FID score (15.13) compared to Male → Female (21.73). This indicates that Female → Male transformations produce more visually realistic and domain-consistent images.

5) StarGAN v2: StarGAN v2 uses two methods for evaluation: latent-guided synthesis and reference-guided synthesis. For latent-guided synthesis, we use 10 latent vectors to translate a source image to the target domain. For reference-guided synthesis, we use 10 reference images to translate a source image to the target domain. Evaluating with the former means assessing the quality of transformations with random style codes. Evaluating with the latter means assessing the quality of transferring style attributes accurately when explicitly given reference images while transforming to the target domain.

- **Cycle Consistency Error:** Although cycle consistency error was computed, it cannot be compared to the cycle consistency errors found for CycleGAN because the mechanism for computing it is different. The cycle consistency error was 0.1914 for latent-guided synthesis and 0.1840 for reference-guided synthesis.
- **Fréchet Inception Distance (FID):** For latent-guided synthesis, the Female → Male transformation achieves a lower FID score (17.36) compared to Male → Female (21.26), suggesting that Female → Male transformations produce more visually realistic and domain-consistent images in the latent space. Similarly, for reference-guided transformations, Female → Male achieves a lower FID score (18.50) compared to the opposite direction (20.98). Reference-guided transformations consistently outperform latent-guided ones, indicating that using reference images provides improved image quality and style fidelity. Overall, the FID scores from both styles of synthesis suggest that Female → Male produces more visually realistic and domain-consistent images.

6) Comparison Between the Models:

- **Cycle Consistency Error:**
  - The **Base CycleGAN** exhibits the highest cycle consistency errors, serving as a benchmark for evaluating subsequent improvements.
  - Introducing the **Self-Attention Mechanism** slightly reduces these errors, indicating moderate improvement in reconstruction accuracy.
  - The **Spatial-Attention Mechanism** achieves even lower cycle consistency errors compared to the self-attention mechanism, demonstrating better reconstruction performance.
  - The **U-Net Modifications** yield the lowest cycle consistency errors among the CycleGAN variants, highlighting its enhanced reconstruction capabilities.
- **Identity Preservation Error:**
  - The **Base CycleGAN** performs moderately well, with slightly better identity retention for Female → Male transformations.
  - The **Self-Attention Mechanism** shows a slight decline in identity preservation compared to the base model, particularly for Male → Female transformations.
  - The **Spatial-Attention Mechanism** achieves the best identity preservation performance, particularly excelling in Male → Female transformations.
  - The **U-Net Modifications** maintain competitive identity preservation, with slightly higher errors compared to the spatial-attention mechanism.
- **Fréchet Inception Distance (FID):**
  - The **Base CycleGAN** achieves reasonable FID scores with minimal directional bias in visual realism.
  - The **Self-Attention Mechanism** has the worst FID scores and introduces an imbalance in transformation quality, with an FID score 7.13 points higher for Male → Female transformations than Female → Male.
  - The **Spatial-Attention Mechanism** produces improvements in visual realism compared to the baseline, particularly for Female → Male transformations, achieving the lowest FID scores among the CycleGAN vari-

ants. The imbalance in transformation quality remains, with an FID score 1.72 points higher for Male → Female transformations.
  - The **U-Net Modifications** maintain strong performance in Female → Male transformations, although FID scores for Male → Female transformations are slightly higher than those of the spatial-attention mechanism. The imbalance in transformation quality remains, with an FID score 6.6 points higher for Male → Female transformations.
  - For **StarGAN v2**, both latent- and reference-guided synthesis achieve competitive FID scores, with Female → Male transformations consistently outperforming Male → Female transformations. Reference-guided synthesis offers improved image quality and style fidelity compared to latent-guided synthesis.
- **Overall Performance:**
  - The **Base CycleGAN** provides a reliable baseline but lacks the refinements of the enhanced models.
  - The **Self-Attention Mechanism** offers improved reconstruction accuracy but at the cost of identity preservation and balanced FID scores.
  - The **Spatial-Attention Mechanism** strikes an effective balance between identity preservation and visual realism, excelling in Female → Male transformations.
  - The **U-Net Modifications** achieve the best reconstruction accuracy while maintaining competitive FID scores, offering a robust solution with slight trade-offs in identity preservation.
  - **StarGAN v2** stands out for its flexibility and high-quality transformations, particularly excelling in reference-guided synthesis and Female → Male transformations.

B. Qualitative

To complement the quantitative analysis, we performed a qualitative evaluation of the transformed images.

1) Base CycleGAN: Figure 2 shows the results for Male → Female and Female → Male transformations using the unmodified CycleGAN model. The top row illustrates some of the better transformations, while the bottom row demonstrates poor results. The ratio of successful to unsuccessful transformations was approximately 1:1, highlighting the limitations of the base model and underscoring the need for enhanced architectural modifications.

- **Male → Female Transformations:** In the top row, masculine features such as a prominent jawline and facial hair were effectively softened, resulting in feminine attributes like smoother skin and less angular facial structures. Conversely, in the bottom row, the model failed to sufficiently soften masculine traits, leading to outputs that lacked distinctly feminine characteristics.
- **Female → Male Transformations:** In the top row, feminine features were convincingly masculinized, with successful modifications such as thicker eyebrows, broader jawlines, and coarser skin textures. However, in the bottom row, transformations were less effective, as feminine attributes like softer facial structures and smoother textures were not sufficiently altered, resulting in outputs that failed to align with the target masculine domain.

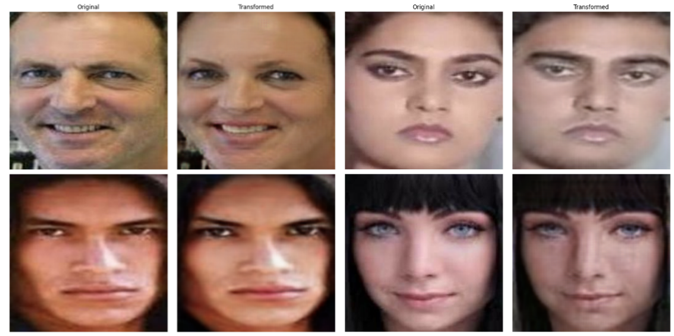

Fig. 2: Base CycleGAN Results. The left panel shows Male → Female transformations, while the right panel shows Female → Male transformations.

2) CycleGAN w/ Self-Attention Mechanism: Figure 3 showcases the results for Male → Female and Female → Male transformations using the self-attention mechanism.

- **Male → Female Transformations:** The model effectively softens facial features and adds feminine attributes, as seen in Figure 3. Examples include rounder and larger eyes, thinner eyebrows, a shorter and rounder nose, and makeup effects such as red lips and eyeliner. Additionally, the removal of masculine traits such as a mustache enhances the transformation. However, some artifacts, such as exaggerated makeup, occasionally reduce the naturalness of the output.
- **Female → Male Transformations:** Transformations in this direction appear more realistic, with the model simulating broader facial structures, rougher skin textures, and subtler masculine attributes. These results are consistent with the lower FID score observed in the quantitative analysis, further confirming the superiority of the Female → Male transformations over Male → Female.

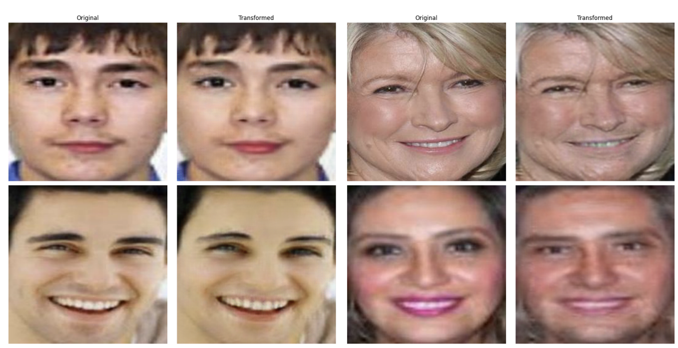

Fig. 3: CycleGAN w/ Self-Attention Mechanism Test Results. The left panel shows Male → Female transformations, while the right panel shows Female → Male transformations.

3) CycleGAN w/ Spatial-Attention Mechanism: Figure 4 showcases the results for Male → Female and Female → Male transformations using the spatial-attention mechanism.

- **Male → Female Transformations:** The model effectively softens facial features and adds traditionally feminine attributes, as seen in Figure 4. Similar to the self-attention extension, we see more emphasized eyes, thinner eyebrows, and makeup effects such as smoothing skin, eyeliner, and reddening of the lips. We also see a general brightening of the skin.
- **Female → Male Transformations:** While the facial structure remains the same, we see deepening of facial shadows, muting of coloration, and more pronounced eyebrows. We also see the faint addition of facial hair. Similar to the self-attention extension, we see a lower FID score for Female → Male (14.76) transformation in comparison to the Male → Female transformation (16.48).

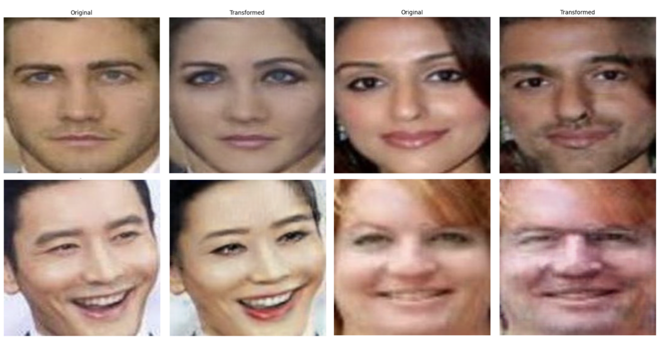

Fig. 4: CycleGAN w/ Spatial-Attention Mechanism Test Results. The left panel shows Male → Female transformations, while the right panel shows Female → Male transformations.

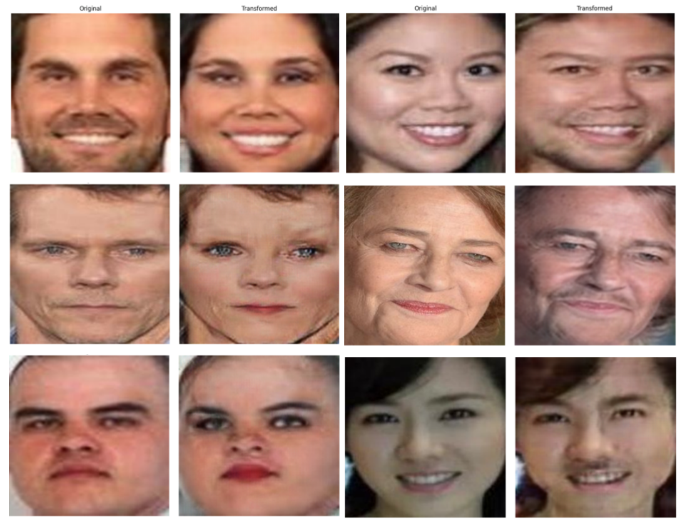

Fig. 5: CycleGAN w/ U-Net Modification Test Results. The left panel shows Male → Female transformations, while the right panel shows Female → Male transformations.

4) CycleGAN w/ U-Net Modification: Figure 5 showcases the results for Male → Female and Female → Male transformations using the self-attention mechanism.

- **Male → Female Transformations:** In this direction of transformation, eyebrows are thinner, eyes are larger and have eyeshadow, nasion is less depressed, cheeks are slightly more blush, facial hair is removed, and lips are thicker and rosier.
- **Female → Male Transformations:** Conversely, eyebrows are thicker, eyes are narrower, facial hair is added, and lips are thinner and paler. The implication from the FID scores that Female → Male transformations produce more visually realistic and domain-consistent image is visually apparent.

5) StarGAN v2: Figure 6 demonstrates the StarGAN v2 transformations. Each row is styled according to the reference image in the first column, which defines the ”style” for the entire row. For example, in the first row, a female reference image is applied to female source images to enhance feminine characteristics. Similarly, in the fourth row, a male reference image transforms male source images into a more masculine style.

- **Male → Female Transformations:** Like the other models, when applying this transformation (in this case by using a female reference image on a male source image), the males appear to have more feminine features. For example, in the first row, there are 5 male source images, and applying the reference image of the lady in the second row's first column, we see that the men have paler faces, like the reference image's lady, and might have smoother faces.
- **Female → Male Transformations:** The transformations in this direction seem to make the women more masculine. However, some feminine features persist. While the FID score suggests this direction performs better than the opposite direction, the visual results indicate areas where improvements could be made. Although StarGAN v2 serves as a strong baseline for comparison, it is clear that it still has its flaws, especially given that significantly fewer iterations were allocated for this model's training.

V. Discussion

Through our exploration of CycleGAN architectural modifications, we discovered several interesting patterns in how different attention mechanisms affect gender transformation tasks. Our analysis begins with the base CycleGAN model, which achieved FID scores of 16.87 and 16.44 for Male → Female and Female → Male transformations respectively. These baseline metrics demonstrate that even the unmodified CycleGAN architecture can produce reasonably good results, with nearly balanced performance across both transformation directions. The base model's cycle consistency errors (0.0751 for M→F, 0.0752 for F→M) suggest robust bidirectional mapping capabilities, while its identity preservation errors

Fig. 6: StarGAN v2 Test Results. Each row is styled based on the reference image in its first column. Male source images are transformed into female appearances when styled by a female reference image, and vice versa, illustrating gender transformations guided by the reference image.

(0.0702 for M→F, 0.0669 for F→M) indicate slightly better performance in maintaining facial identity during Female → Male transformations.

Our most surprising finding was that spatial attention significantly outperformed both the baseline and self-attention variants, achieving improved FID scores of 16.48 and 14.76 for Male → Female and Female → Male transformations respectively. This represents a notable improvement over the base model's FID scores, particularly for Female → Male transformations where we observed a 10.2% reduction in FID score (from 16.44 to 14.76). We initially hypothesized that self-attention would perform better due to its ability to model global relationships, but our results suggest that focusing on specific facial regions is more effective for gender transformation. This makes intuitive sense in retrospect, as gender-specific features tend to be localized in particular areas of the face.

One of the most fascinating aspects of our results was the consistent asymmetry between Male → Female and Female → Male transformations across all our architectural variants, starting with the base model and persisting through our modifications. The base model showed a slight preference for Female → Male transformations (FID 16.44 vs 16.87), a pattern that became more pronounced with our architectural modifications. While working with the dataset, we noticed that female facial features often involve more complex elements like varying makeup styles and hairstyles, which might explain why the Male → Female direction proved more challenging. We also suspect that inherent biases in our training dataset might have influenced this directional performance difference, though further investigation would be needed to confirm this hypothesis.

Our comparison with StarGAN v2 yielded some unexpected insights. While StarGAN v2 achieved comparable results with only 1.875% of the images seen by the other models, its training process was more computationally intensive due to its architectural complexity. This highlights an interesting trade-off between data efficiency and computational requirements. The base CycleGAN's simpler architecture, with its straightforward cycle consistency approach, provides a foundation that can be enhanced through targeted modifications without incurring the computational overhead of more complex models.

Our U-Net modifications, in particular, managed to reduce the parameter count by 26.2% while maintaining competitive performance relative to the base model's metrics. We found this especially exciting as it suggests that thoughtful architectural changes can be just as effective as more complex solutions. The modified U-Net architecture achieved FID scores of 21.73 and 15.13 for Male → Female and Female → Male transformations respectively, showing that simplification doesn't necessarily come at the cost of performance.

Looking ahead, we see several promising directions for future work. While our modifications improved overall quality compared to the base model, we still haven't matched StarGAN v2's ability to control fine-grained style features. We're particularly interested in exploring ways to incorporate style-based features while keeping the architecture relatively simple. The base CycleGAN's performance suggests that there's still untapped potential in the fundamental architecture that could be unleashed through careful modification.

We also think it would be valuable to test these modifications on other image-to-image translation tasks to better understand their generalizability beyond gender transformation. Additionally, during our research, we frequently discussed the importance of human perception versus quantitative metrics like FID scores and cycle consistency errors. We believe a systematic human evaluation study would provide valuable insights that our current metrics might be missing, particularly given the nuanced nature of gender presentation in faces.

Our work demonstrates that sometimes simpler solutions, when carefully designed with domain knowledge in mind, can compete with more complex architectures. The base CycleGAN's solid fundamental performance (FID scores around 16.5) provided a strong foundation for targeted improvements, suggesting that architectural complexity isn't always necessary for achieving good results in image-to-image translation tasks. We hope these findings will encourage more research into targeted architectural modifications as an alternative to increasing model complexity.

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